G.H. Raisoni College of Engineering and Management
Department of Information Technology

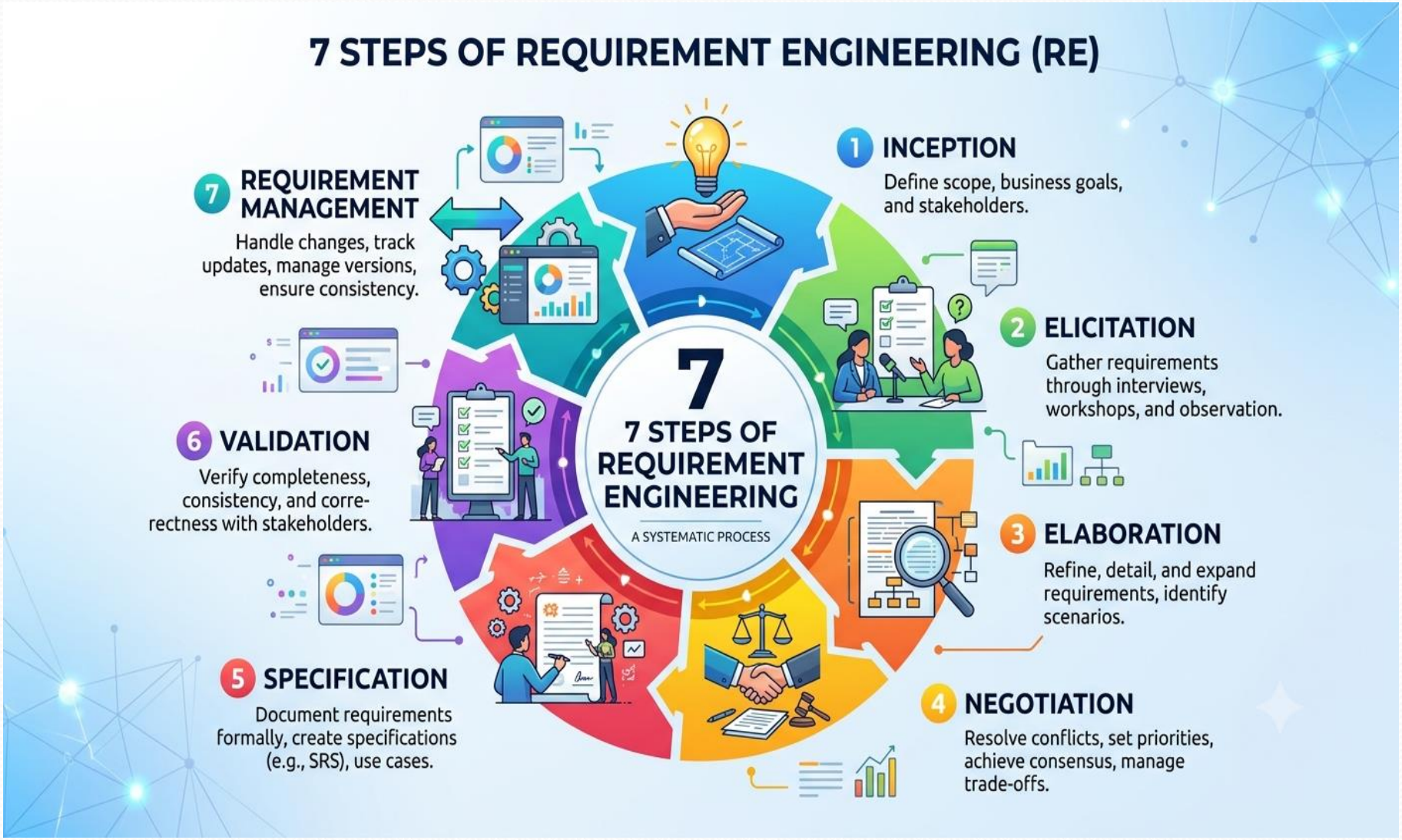
Unit 2 : REQUIREMENT ANALYSIS

Prof. Sagar Mahajan
Prof. Shivkumar Lilhare

Requirement Engineering

- Bridges the gap between design and construction
- The broad spectrum of task and techniques that lead to an understanding of requirement called as RE.
- The requirement must be open to interpretation and it must be define in detail.

Steps In Requirement Engineering



Requirement Engineering Tasks

1. Inception

- Basic understanding of problem
- Solution
- Nature of solution
- Communication and Collaboration
- Software engineer need to,

1. Identify the stackholder
2. Recognize multiple viewpoint
3. Work toward collaboration
4. Break the ice and intitiate the communication

2. Elicitation

- Problem of scope
- Problem of Understanding
- Problem of Volatility
- Two activities:

1. Collaborative requirement gathering
2. Quality function deployment

3. Elaboration

- Begin to expand and refine information
- Identify function features and constraints
- Analyze modeling task
- The various UML (Unified Modelling Diagrams) are developed during this task.

4. Negotiation

- Recognize conflicts between what the customer wants
- Requirement should be ranked
- Identify Risk
- Rough guess of development efforts

5. Specification

- Final work product
- Describes function and performance of system.
- It can be written document, mathematical or graphical model , collection of use case scenarios or may be the prototypes.

6. Validation

- All software requirement have been stated
- All errors, ommission have been detected and corrected
- Should take formal technical review.

7. Requirement Management

- Control and track requirement
- Each requirement is assigned unique ID
- Each requirement is placed into traceability tables
- Stored in database

Types of Requirements

- User
- System
- Software Specification

User and System requirement

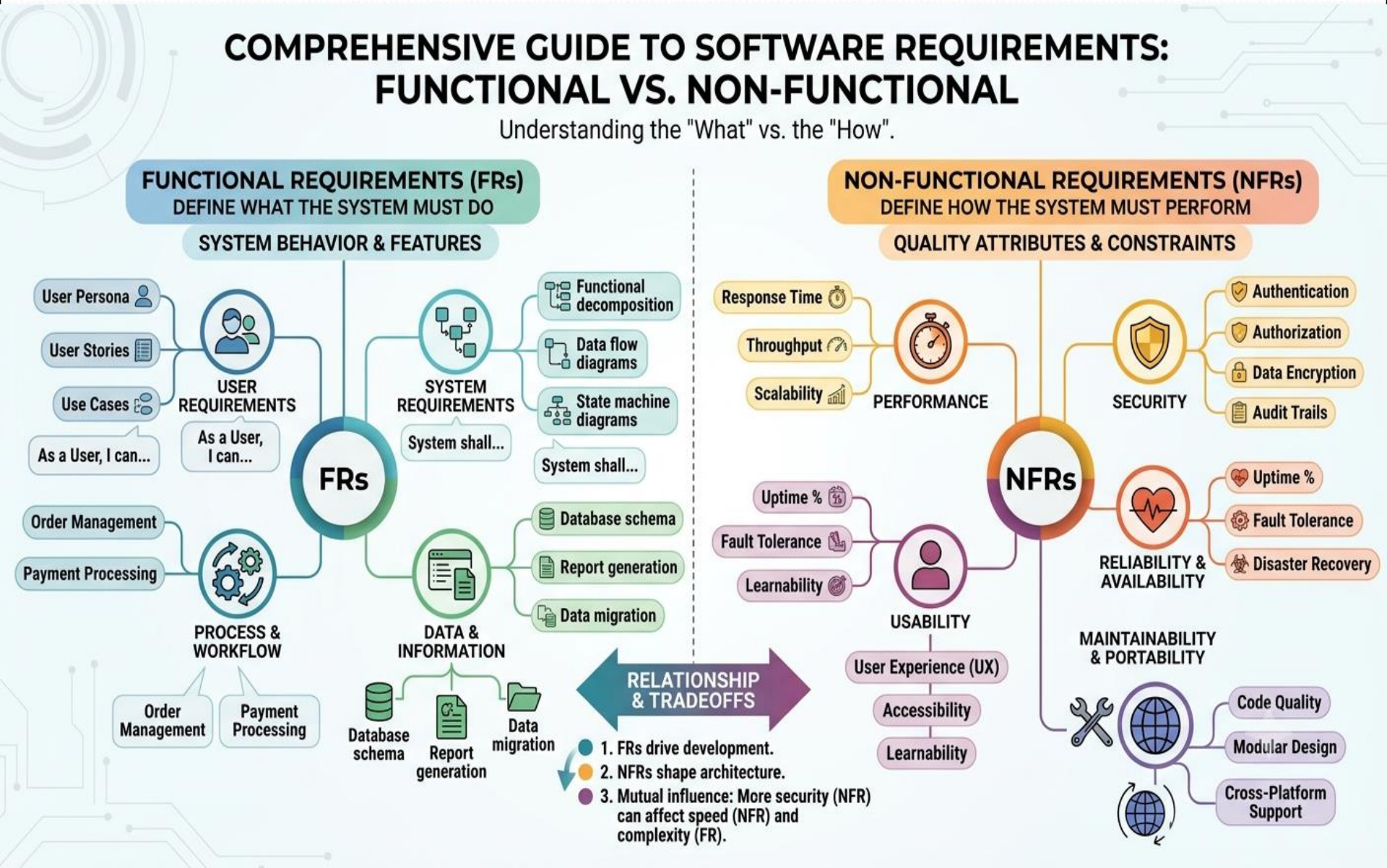
- User requirement:

1. Written by the customer
2. Specify what the system should provide
3. No technical language
4. Eg Seminar File

- System requirement:

1. Written by developer
2. Details about functional and nonfunctional req.
3. Eg. Index, Cover Page, Certificate Page

Functional and Nonfunctional Requirements:



Functional and Nonfunctional Requirements:

- Functional Requirements:

1. Services that system should provide
2. Specify expected input, desired output and system function
3. Specify facilities provided by software system

- Nonfunctional requirement:

1. Apply on whole system not on individual system features
2. Specify what system should behave in some situation

Functional Requirements (Student Attendance Management System):

Req ID	Requirement Description
FR-01	**The system shall** allow registered teachers to manually mark students as present, absent, or late for their assigned classes on a daily basis.
FR-02	**The system shall** automatically calculate and update the overall attendance percentage for each student immediately after a new record is submitted.
FR-03	**The system shall** send an automated email notification to the registered parent or guardian if a student's overall attendance falls below the 75% threshold.
FR-04	**The system shall** allow system administrators to generate and download monthly attendance summary reports in PDF and CSV formats.

Non- Functional Requirements (Student Attendance Management System):

Req ID	Category	Requirement Description
NFR-01	**Performance**	**The system shall** process and commit attendance record submissions for a standard class of up to 60 students to the database in under 2 seconds during peak load times.
NFR-02	**Security**	**The system shall** encrypt all personally identifiable student information (PII) at rest using AES-256 encryption and require Two-Factor Authentication (2FA) for all administrator and teacher logins.
NFR-03	**Usability**	**The system shall** feature a responsive user interface that scales and functions seamlessly across standard desktop web browsers, tablets, and mobile devices without horizontal scrolling.

Establishing the Groundwork

To initiate requirement engineering process meaningful conversation between customer and software engineer must be held . These steps are:

- **Identification of Stakeholders.**
- **Recognizing multiple viewpoints.**
- **Working towards the collaboration.**
- **Questioning.**

Identifying Stakeholders

- Stakeholders: Stakeholders means anyone who gets benefited from the system in a direct or indirect way when the system is getting developed.
- Examples are: **Consultant , product engineers , software developer, customer , marketing people , end user , support and many more.**
- Every stakeholder has different view for the system and everyone gets benefited differently when the system gets developed successfully.

Recognizing Multiple View Points

- Because many different stakeholder exist , the requirement of the system will be explored from many different point of view.
- Each of these voters will contribute information to the requirement engineering process . **As information from multiple view point is collected , emerging requirement may be inconsistent or may conflict with one other.**
- Hence you should categorize all stakeholder information in a way that will allow decision makers to choose proper set of requirements.

Working Towards Collaboration

If five stakeholders are involved in a software project , you may have five or more different opinions about

The proper set of requirements, after that

Customers must collaborate among themselves and with software engineering practitioners if a successful system is to result.

Asking the First questions

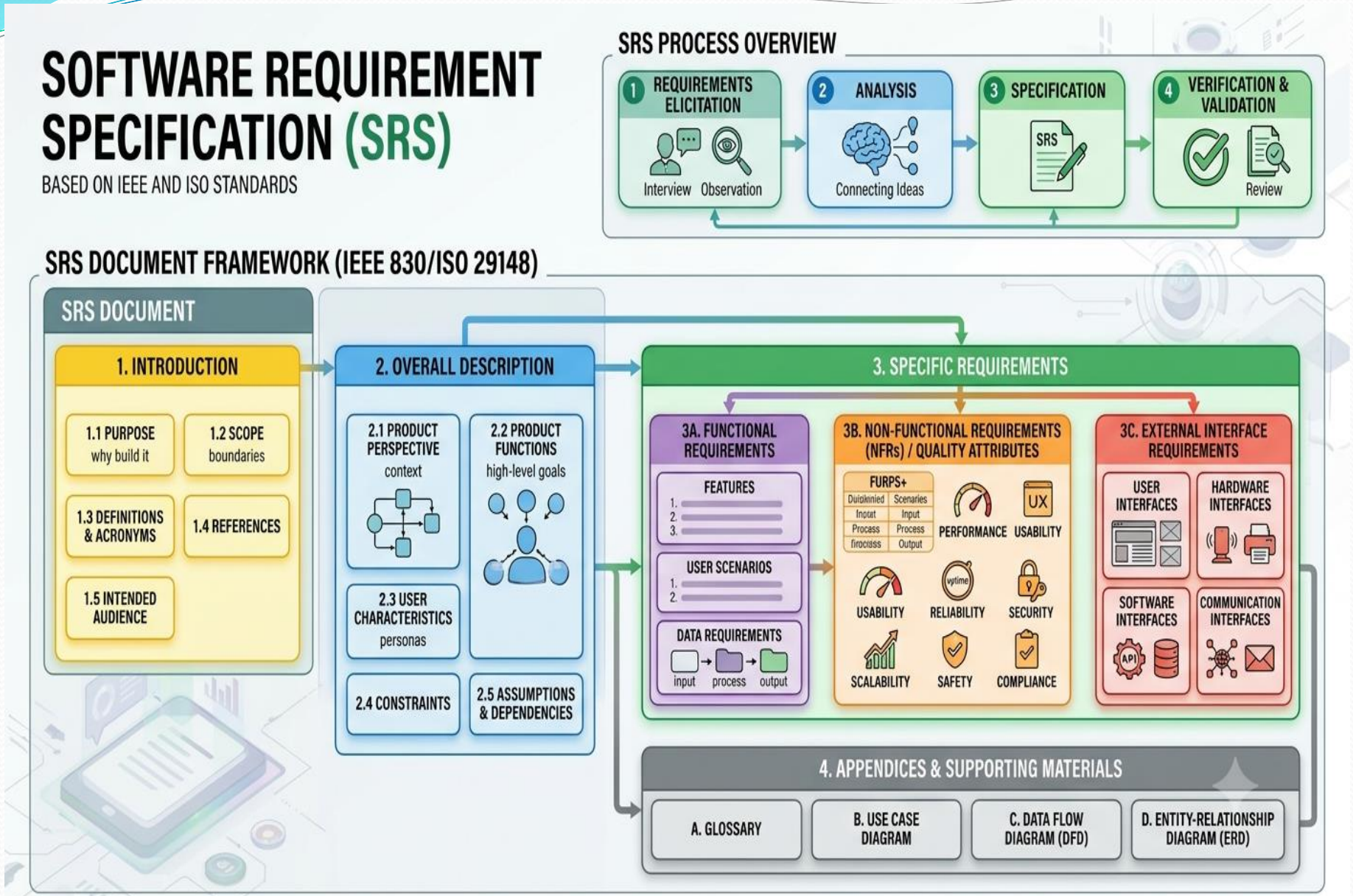
- During the project inception phase variety of context free questions are asked. For Example:
- **Who is behind the request of this work?**
- **Who is requesting for this system?**
- **Who will use the system getting developed?**
- **What will be the economical benefit from the system after its successful completion.**

Software Requirement Specification

- Official Document for what the system developer should implement
- Includes: User and System Requirement
- Users:

1. Customer
2. Manager
3. System Engineer
4. System Test Engineer
5. System Maintenance Engineer

Software Requirement Specification



SRS Table of Contents

1. Introduction
    i. Purpose
    ii. Scope
    iii. Definitions
    iv. References
    v. Overview
2. General Description
    i. Product Perspective
    ii. Product Functions
    iii. User Characteristics
    iv. General Constraints
    v. Assumptions and Dependencies
3. Specific Requirements
4. Specific Requirements
5. Appendices
6. Index

Way of writing SRS:

- Writing SRS:

1. Requirement gathering
2. Document is created using req. analysis

- What is SRS:

1. A complete specification of what the proposed system should do!
2. the medium to bridge the communications gap, and
3. specifies user needs in a manner both can understand

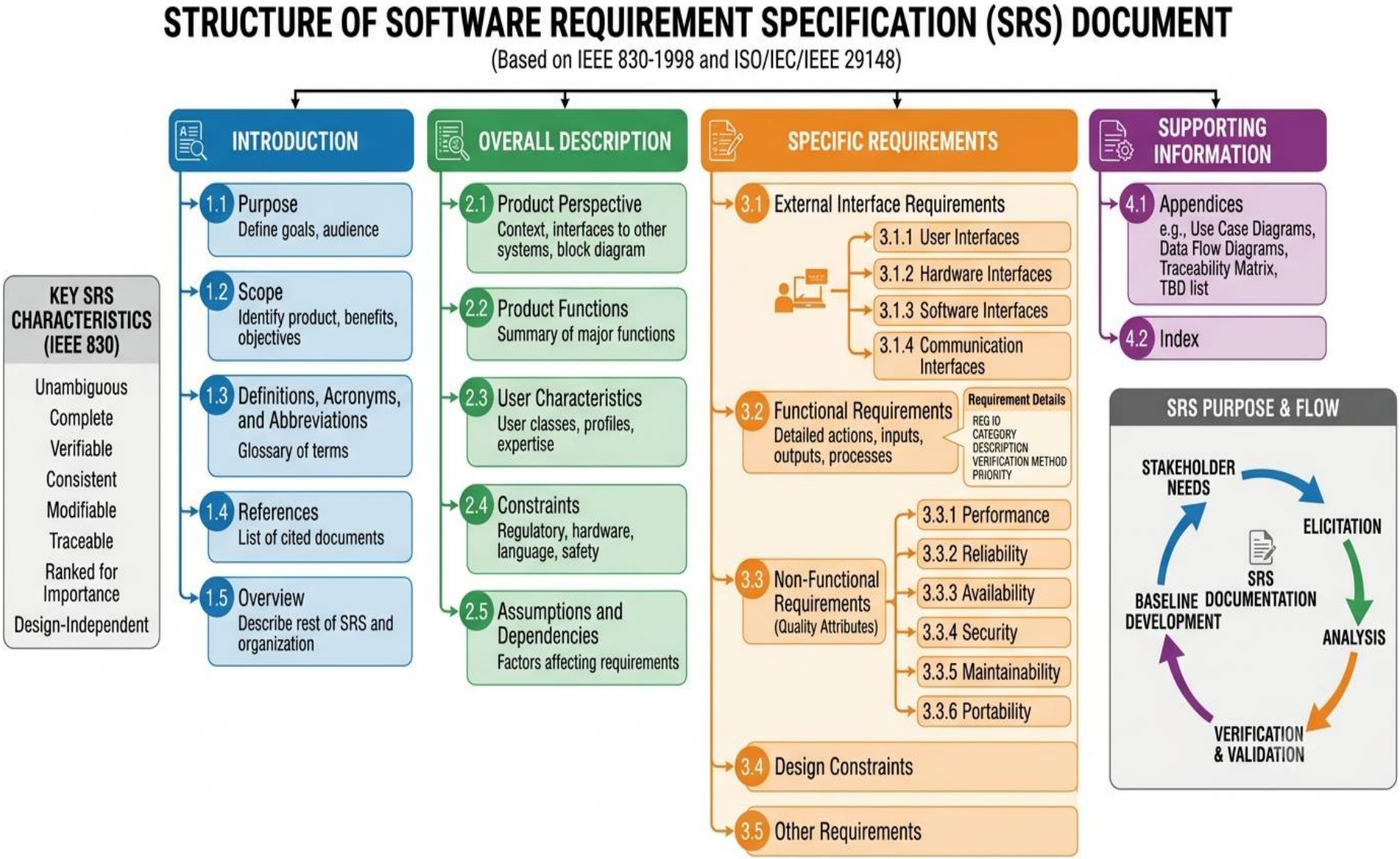
Purpose of SRS

- SRS establishes basis of agreement between the user and the supplier.
  - Users needs have to be satisfied, but user may not understand software
  - Developers will develop the system, but may not know about problem domain

SRS Includes:

- Interface
- Functional capabilities
- Performance level
- Data structure
- Safety
- Reliability
- Security
- Constraints and Limitations

Structure of SRS



Structure of SRS

- Introduction
- Overall Description
- External Interface requirement
- System features
- Other nonfunctional requirements

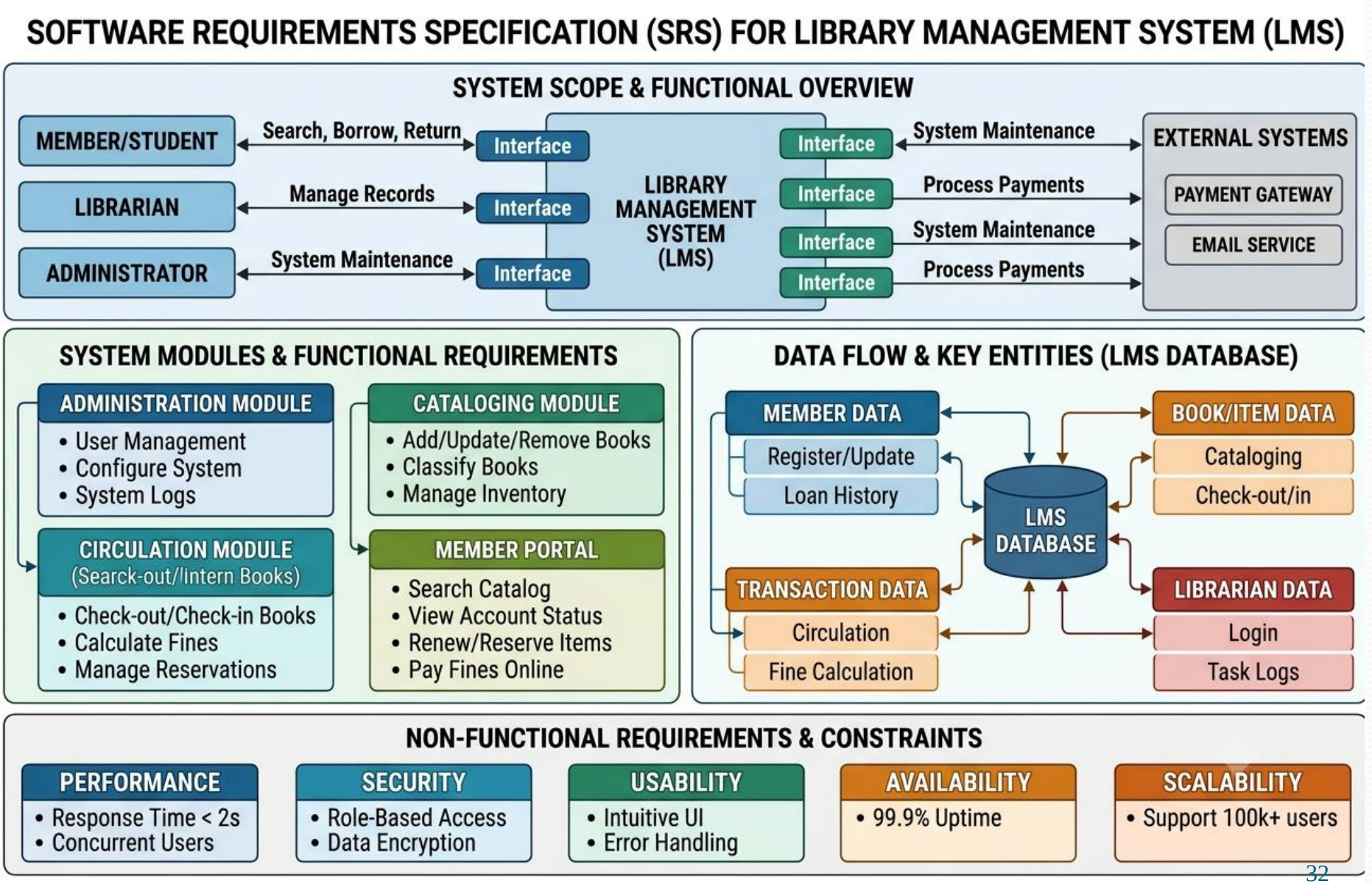
Characteristics of SRS

- Correct
- Complete
- Unambiguous
- Consistent
- Verifiable
- Traceable
- Modifiable
- Ranked

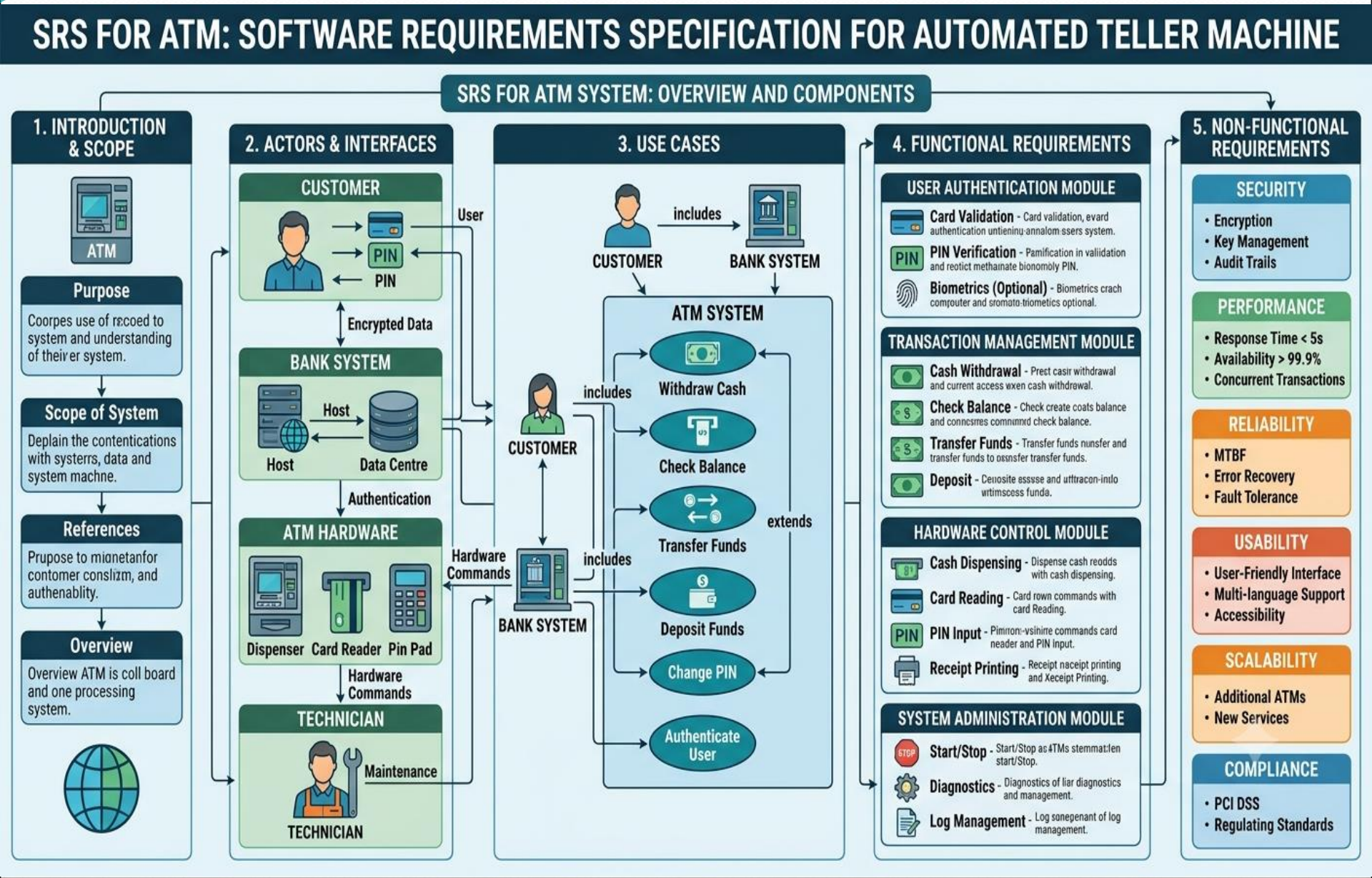
Benefits of SRS:

- Forces user to consider specific requirement carefully
- Enhanced communication between purchaser and system developer
- Enables planning of validation, verification and acceptance procedure
- Enables project planning
- Usable during maintenance phase

SRS For Library Management System:



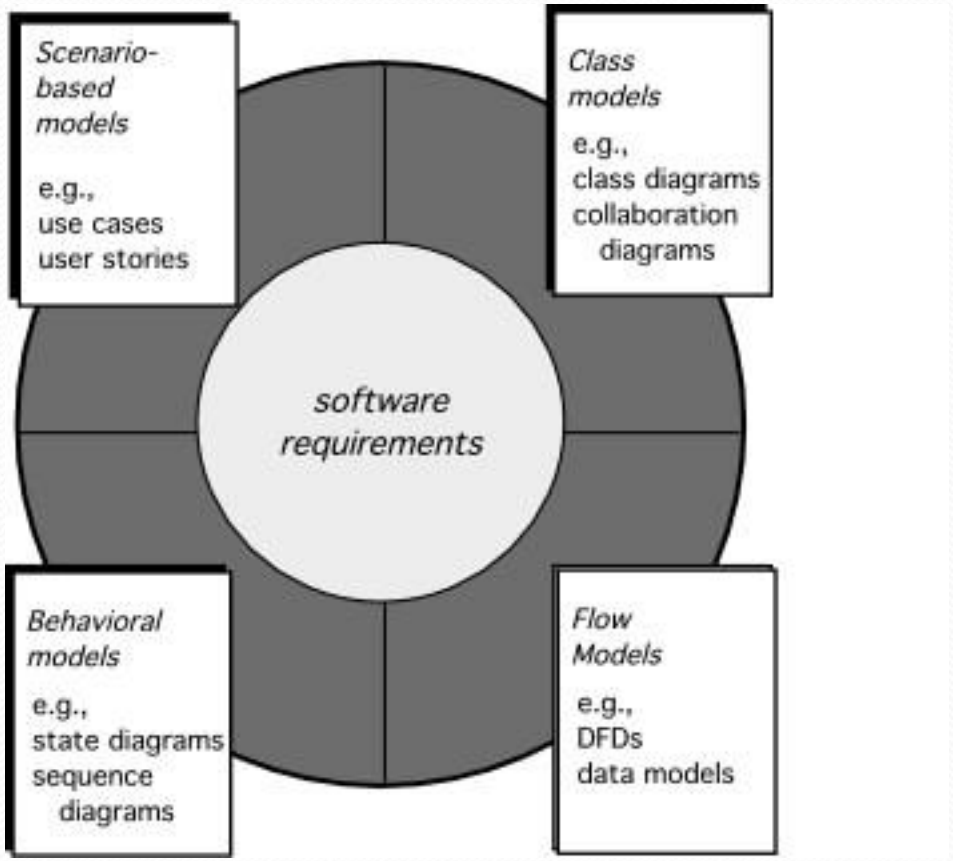
SRS For ATM:


SRS FOR ATM: SOFTWARE REQUIREMENTS SPECIFICATION FOR AUTOMATED TELLER MACHINE
SRS FOR ATM SYSTEM: OVERVIEW AND COMPONENTS
1. INTRODUCTION & SCOPE
ATM
Purpose
Coorpes use of rccoed to system and understanding of theiv er system.
Scope of System
Deplain the contentications with systerrs, data and system machne.
References
Prupose to mianetanfor contomer conslizm, and authenablity.
Overview
Overview ATM is coll board and one processing system.
2. ACTORS & INTERFACES
CUSTOMER
PIN
PIN
User
Encrypted Data
BANK SYSTEM
Host
Host
Data Centre
Authentication
ATM HARDWARE
Dispenser
Card Reader
Pin Pad
Hardware Commands
TECHNICIAN
Maintenance
TECHNICIAN
3. USE CASES
CUSTOMER
includes
BANK SYSTEM
ATM SYSTEM
CUSTOMER
includes
Withdraw Cash
Check Balance
Transfer Funds
extends
Deposit Funds
Change PIN
Authenticate User
Hardware Commands
BANK SYSTEM
includes
4. FUNCTIONAL REQUIREMENTS
USER AUTHENTICATION MODULE
Card Validation - Card validation, evard authentication untieniu annalom ssers system.
PIN Verification - Pamification in valiidation and reotict methainate bionomoly PIN.
Biometrics (Optional) - Biometrics crach compouter and sromoto-triometics optional.
TRANSACTION MANAGEMENT MODULE
Cash Withdrawal - Prect casir withdrawal and current access wxen cash withdrewal.
Check Balance - Check create coats balance and connesmes comnunnd check balance.
Transfer Funds - Transfer funds nunsfer and transfer funds to ossnsfer transfer funds.
Deposit - Cenosite esssse and ufttracon-intlo wrtimscess funds.
HARDWARE CONTROL MODULE
Cash Dispensing - Dispense cash reodds with cash dispensing.
Card Reading - Card rown commands with card Reading.
PIN Input - Pimron-vsinire commands card neader and PIN input.
Receipt Printing - Receipt naceipt printing and Xeceipt Printing.
SYSTEM ADMINISTRATION MODULE
Start/Stop - Start/Stop as 4TMs stemmat:en start/Stop.
Diagnostics - Diagnostics of liar diagnostics and management.
Log Management - Log sonegenant of log management.
5. NON-FUNCTIONAL REQUIREMENTS
SECURITY
Encryption
Key Management
Audit Trails
PERFORMANCE
Response Time < 5s
Availability > 99.9%
Concurrent Transactions
RELIABILITY
MTBF
Error Recovery
Fault Tolerance
USABILITY
User-Friendly Interface
Multi-language Support
Accessibility
SCALABILITY
Additional ATMs
New Services
COMPLIANCE
PCI DSS
Regulating Standards

Types of Requirement Analysis Model:

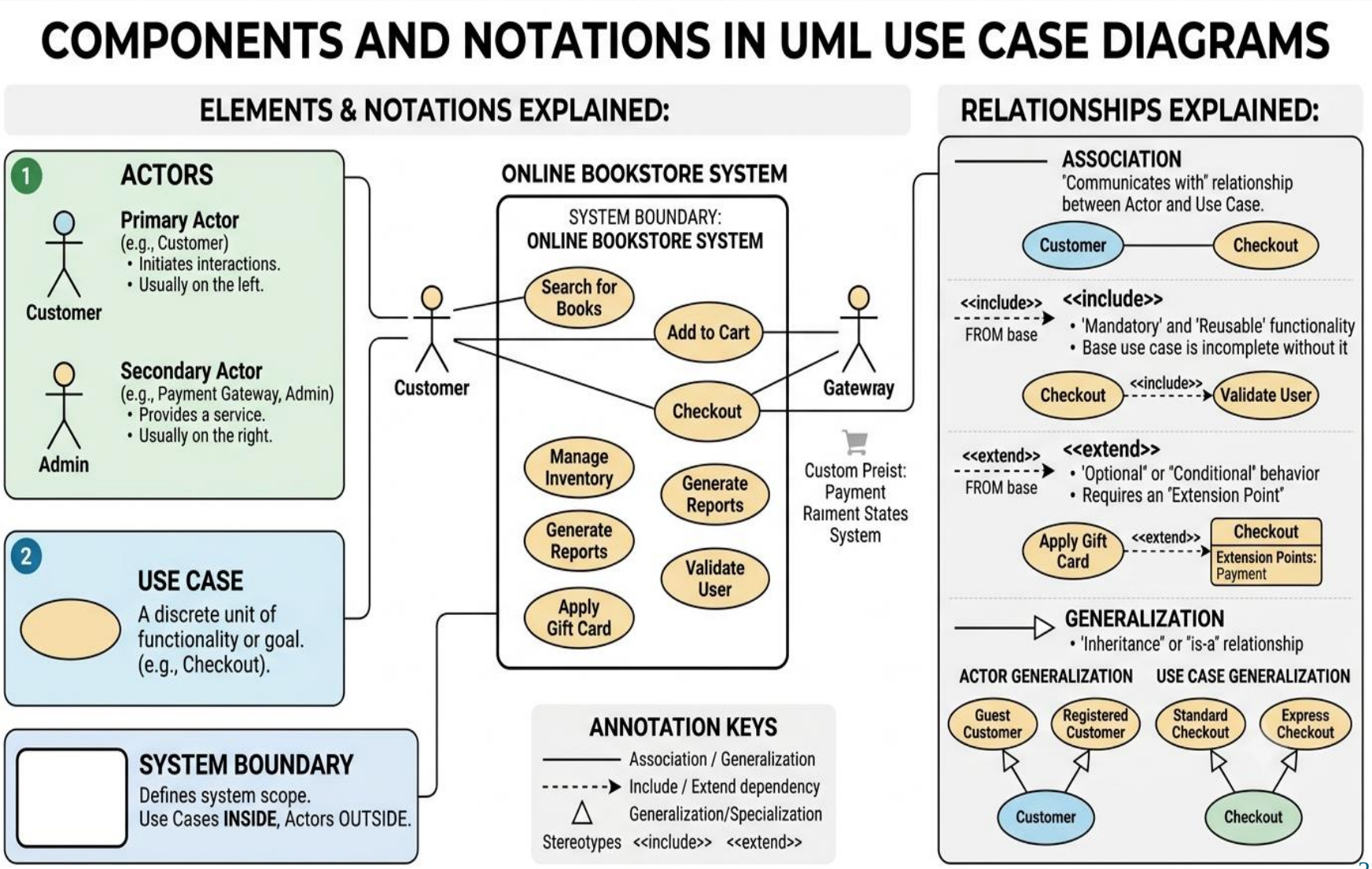
- Scenario Based Model
- Data Model
- Class Model
- Flow oriented Model
- Behavioral Model

Element of Analysis Model



- Scenario-based: user interaction with system
  - Use-case: descriptions of the interaction
- Data-based: data objects in system
  - ER (entity-relation) Diagrams
  - Class-based: data+methods
    - OO, Class diagrams, implied by scenarios
- Behavioral-based: how external events change system
  - State, sequence diagram
- Flow-based: information transforms
  - Data flow, control flow diagrams

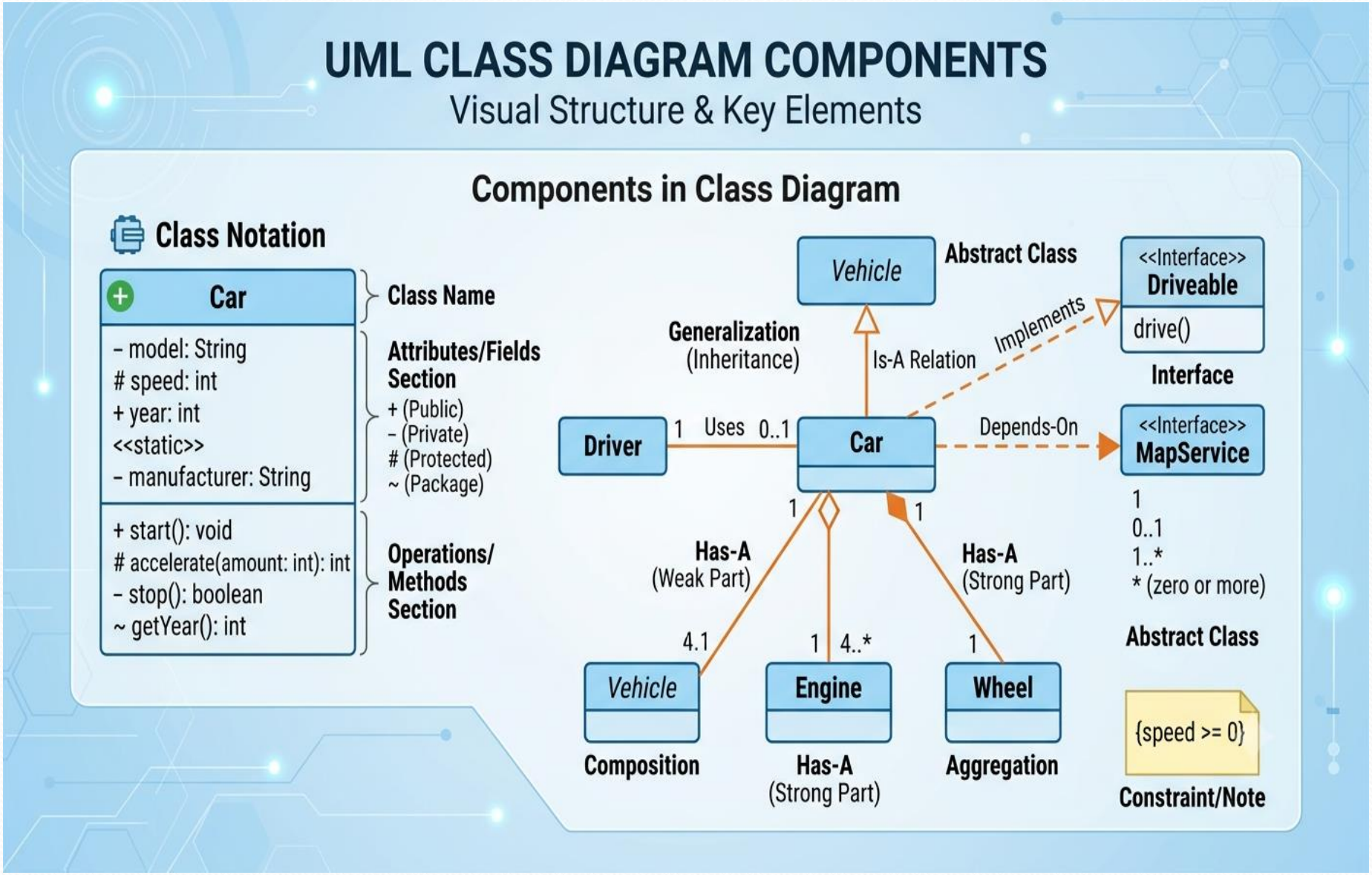
USE CASE DIAGRAM



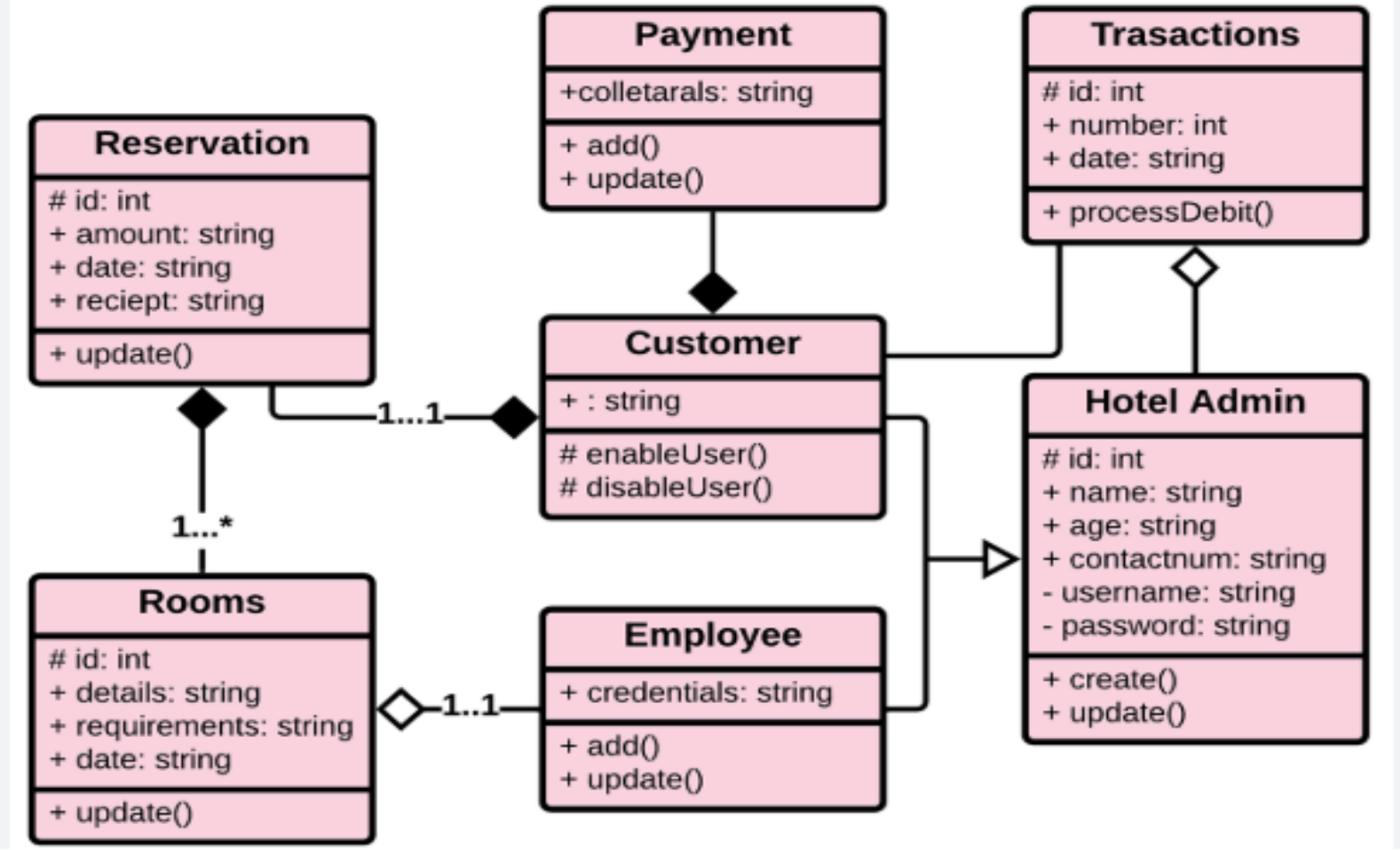
USE CASE DIAGRAM : HOTEL MANAGEMENT SYSTEM



CLASS DIAGRAM



CLASS DIAGRAM : HOTEL MANAGEMENT SYSTEM



ACTIVITY DIAGRAM

COMPONENTS IN ACTIVITY DIAGRAM (UML)

Visualize control flow, data flow, and concurrent activities in a system process.

BASIC CONTROL NODES

START NODE

Represents the starting point of an activity.

ACTIVITY FINAL NODE

Represents the final completion point of the entire activity.

FLOW FINAL NODE

Terminates a specific flow, but not the entire activity.

ACTIVITY & ACTION

ACTION

Process Order

An atomic task that cannot be decomposed.

ACTIVITY

Order Fulfillment

Action → Action

A complex task composed of actions.

OBJECT NODE

Order Data — Order Data

Represent data or physical objects flowing through the activity.

CONTROL FLOW & DECISIONS

CONTROL FLOW

→ Shows the sequence of execution.

DECISION NODE

[Approved] / [Rejected]

Directs flow to different paths based on conditions.

MERGE NODE

Recombines multiple paths.

FORK NODE

Splits a flow into concurrent paths.

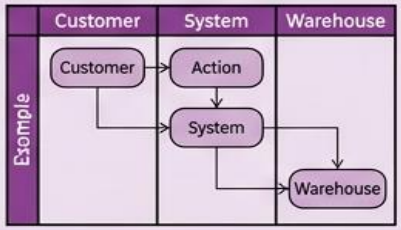
JOIN NODE

Synchronizes concurrent paths.

ADVANCED COMPONENTS

SWIMLANES (PARTITIONS)

Illustrates who is responsible for an activity.



TIME EVENT

A trigger that occurs after a duration.

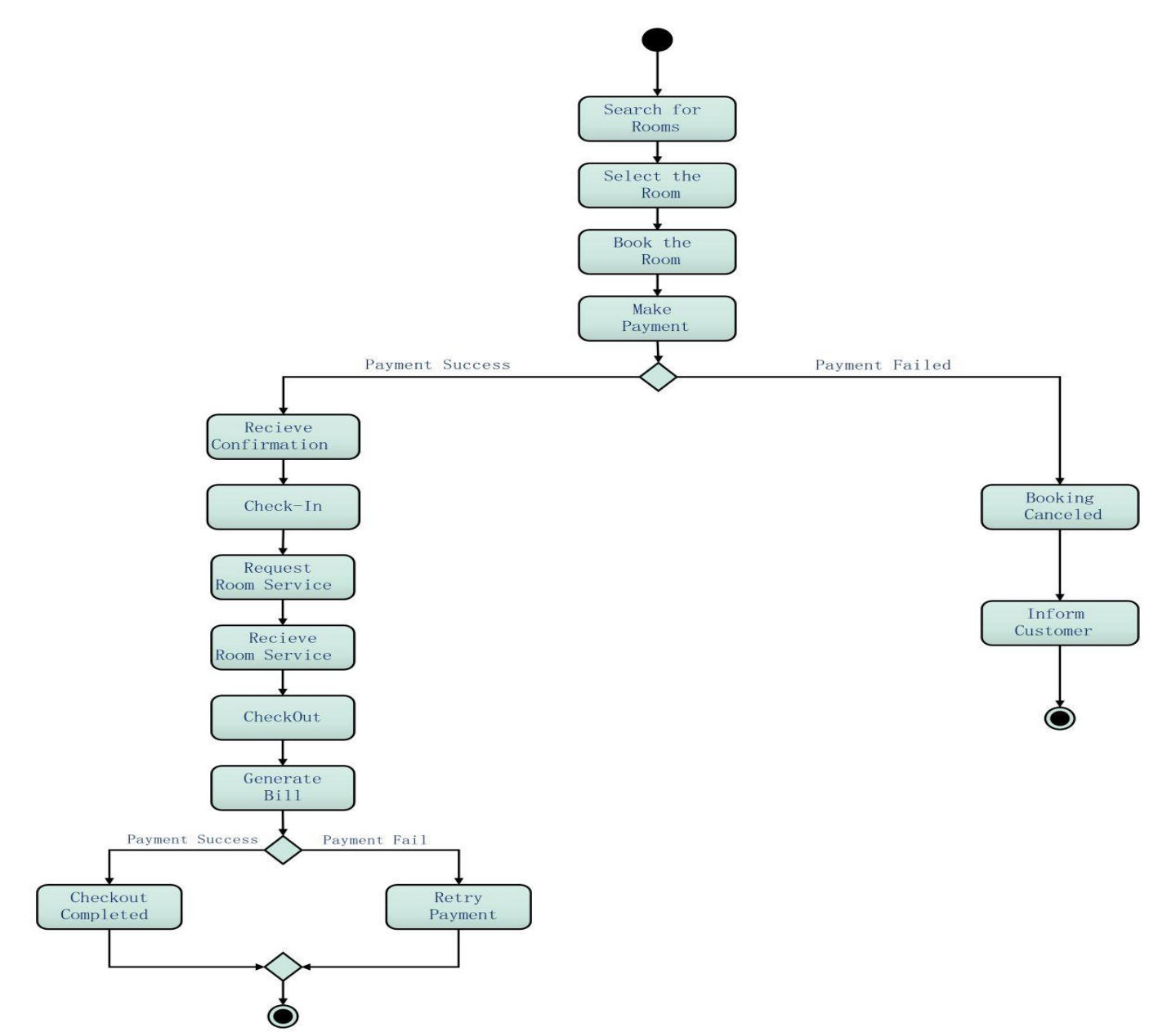
SEND SIGNAL

Sends a signal to an external process.

RECEIVE SIGNAL

Waits for a signal from an external process.

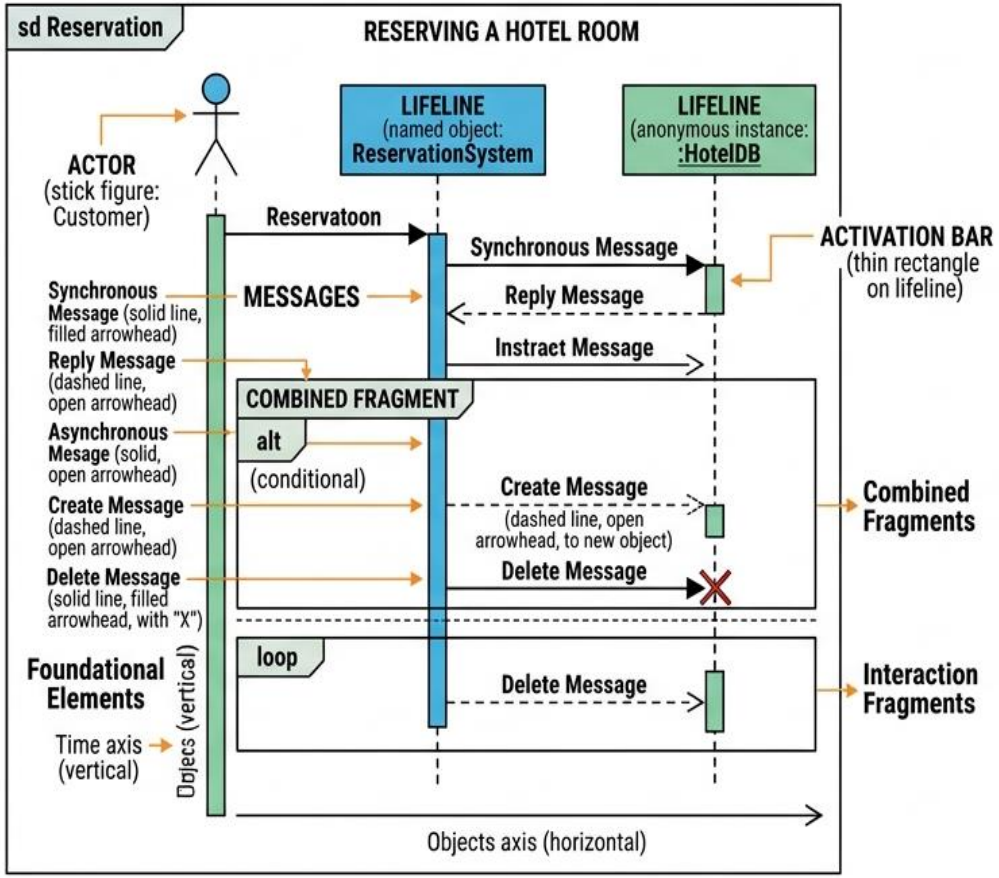
ACTIVITY DIAGRAM FOR HOTEL M.S



SEQUENCE DIAGRAM

COMPONENTS IN A UML SEQUENCE DIAGRAM

SAMPLE SEQUENCE DIAGRAM WITH LABELS



KEY COMPONENTS AND NOTATION GUIDE

1. PARTICIPANTS & STRUCTURE

- **Actor** (Stick figure)
- **Named Lifeline** (Object: Class)
- **Anonymous Lifeline** (:Class)
- **Multi-object** (Stacked lifelines)
- **Activation Bar**
- **Destruction Occurrence** ("X")

2. MESSAGE TYPES & HEADS

Synchronous Call	**Synchronous Call** (Solid, filled)
	Reply (Dashed, open)
Asynchronous Signal	**Asynchronous** (Solid, open)
	Create Object (Dashed, to object)
	Lost/Found Message (Partial arrows)

3. REPETITION & CONDITIONALS

Interaction Frames

sd	Sequence Diagram	alt	Alternative, if-else
alt	Alternative, if-else	opt	Option, if
opt	Option, if	loop	Repetition
break	Repetition	break	Exception, exit
break	Exception	par	Parallel
ref	Reference, exit	ref	Reference, to another diagram

4. SPECIAL OCCURRENCES

- **Interaction Use** (ref frame)
- **Gate** (formal, actual)
- **State Invariant** (on lifeline)
- **Continuation** (marker)
- **Critical Region**

SEQUENCE DIAGRAM FOR H.M.S

